Practical 4 Part II CL-III

Dataset:

Starting from the year 2008 hottest and lowest recorded temperature for Pune City.

Weather Data of Pune,Maharashtra,India.(2008-2022)

Uncover Pune's Weather Secrets: Dive into our Comprehensive Dataset

https://www.kaggle.com/datasets/dipakdeshmukh/weather-data-of-punemaharashtraindia2008-2022

Code: WeatherAnalysis.java

```
import java.io.IOException;
import org.apache.hadoop.conf.Configuration;
import org.apache.hadoop.fs.Path;
import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.LongWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Job;
import org.apache.hadoop.mapreduce.Mapper;
import org.apache.hadoop.mapreduce.Reducer;
import org.apache.hadoop.mapreduce.lib.input.FileInputFormat;
import org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;

public class WeatherAnalysis {

    // 1. THE MAPPER: Reads the file line by line
    public static class MyMapper extends Mapper<LongWritable, Text, Text, IntWritable> {
        public void map(LongWritable key, Text value, Context
```

```
context) throws IOException, InterruptedException {

            String row = value.toString();

            // Skip the header row (the names of the columns)
            if (row.contains("date_time")) {
                return;
            }

            // Split the row by comma to get individual colum
ns
            String[] columns = row.split(",");

            // Extract the year (first 4 characters of the da
te_time column at index 0)
            String myYear = columns[0].substring(0, 4);

            // Get the temperature (from index 20) and turn i
t into a number
            int currentTemp = Integer.parseInt(columns[20]);

            // Send (Year, Temperature) to the Reducer
            context.write(new Text(myYear), new IntWritable(c
urrentTemp));
        }
    }

    // 2. THE REDUCER: Finds the Max and Min for each year
    public static class MyReducer extends Reducer<Text, IntWr
itable, Text, Text> {
        public void reduce(Text myYear, Iterable<IntWritable>
allTemps, Context context) throws IOException, InterruptedExc
eption {

            int highest = -999; // Start very low
            int lowest = 999;   // Start very high
```

```
            // Look at every temperature sent for this specif
ic year
            for (IntWritable tempValue : allTemps) {
                int score = tempValue.get();

                if (score > highest) {
                    highest = score;
                }
                if (score < lowest) {
                    lowest = score;
                }
            }

            // Create a nice looking message
            String result = "Hottest: " + highest + "C, Coole
st: " + lowest + "C";

            // Output the final answer for this year
            context.write(myYear, new Text(result));
        }
    }

    // 3. THE DRIVER: This starts the whole engine
    public static void main(String[] args) throws Exception {
        Configuration conf = new Configuration();
        Job myJob = Job.getInstance(conf, "Weather Task");

        myJob.setJarByClass(WeatherAnalysis.class);
        myJob.setMapperClass(MyMapper.class);
        myJob.setReducerClass(MyReducer.class);

        // Tell Hadoop what type of data the Mapper sends out
        myJob.setMapOutputKeyClass(Text.class);
        myJob.setMapOutputValueClass(IntWritable.class);
```

```
        // Set the input and output folders
        FileInputFormat.addInputPath(myJob, new Path(args
[0]));
        FileOutputFormat.setOutputPath(myJob, new Path(args
[1]));

        System.exit(myJob.waitForCompletion(true) ? 0 : 1);
    }
}
```

OUTPUT:

```
student@aids: ~
(.venv) student@aids:~$ start-dfs.sh
Starting namenodes on [localhost]
Starting datanodes
Starting secondary namenodes [aids]
2026-04-17 16:15:43,866 WARN util.NativeCodeLoader: Unable to load native-hadoop
 library for your platform... using builtin-java classes where applicable
(.venv) student@aids:~$ start-yarn.sh
Starting resourcemanager
Starting nodemanagers
(.venv) student@aids:~$ hadoop fs -ls /weather_output
2026-04-17 16:16:02,146 WARN util.NativeCodeLoader: Unable to load native-hadoop
 library for your platform... using builtin-java classes where applicable
Found 2 items
-rw-r--r--   1 student supergroup          0 2026-04-17 16:06 /weather_output/_S
UCCESS
-rw-r--r--   1 student supergroup        480 2026-04-17 16:06 /weather_output/pa
rt-r-00000
(.venv) student@aids:~$ hadoop fs -cat /weather_output/part-r-00000
2026-04-17 16:16:12,783 WARN util.NativeCodeLoader: Unable to load native-hadoop
 library for your platform... using builtin-java classes where applicable
2008    Hottest: 32C, Coolest: 16C
2009    Hottest: 42C, Coolest: 15C
2010    Hottest: 41C, Coolest: 12C
2011    Hottest: 41C, Coolest: 13C
2012    Hottest: 40C, Coolest: 13C
2013    Hottest: 43C, Coolest: 14C
2014    Hottest: 41C, Coolest: 12C
2015    Hottest: 41C, Coolest: 14C
2016    Hottest: 41C, Coolest: 14C
2017    Hottest: 42C, Coolest: 14C
2018    Hottest: 41C, Coolest: 14C
2019    Hottest: 42C, Coolest: 12C
2020    Hottest: 41C, Coolest: 13C
2021    Hottest: 40C, Coolest: 14C
2022    Hottest: 36C, Coolest: 11C
(.venv) student@aids:~$
```